Settimana: 16

Materia: Fisica
Classe: 5F
Data: 21/02/2026

Argomenti:

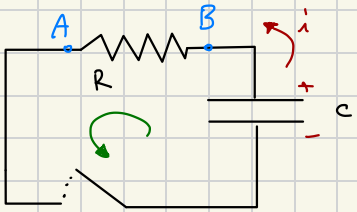
4 esercizi: (1) Condensatori
(2) Circuito I/II Legge di Ohm.
(3) Kirchhoff
(4) Circuito RC

$$Q(t) = C f_{em} \left(1 - e^{-\frac{t}{RC}}\right)$$

$$i(t) = C f_{em} \left(- e^{-\frac{t}{RC}} \left(-\frac{1}{RC}\right)\right)$$

Def: Il tempo caratteristico $t = RC$ misura la rapidità con cui un condensatore di capacità C si carica (o scarica) attraverso un resistore di resistenza R.

Circuito RC - Scarica



- All'inizio tutta la carica si trova sul condensatore
- Una volta chiuso l'interruttore le cariche si ridistribuiscono lungo tutto il circuito

$$C = \frac{Q}{\Delta V}$$

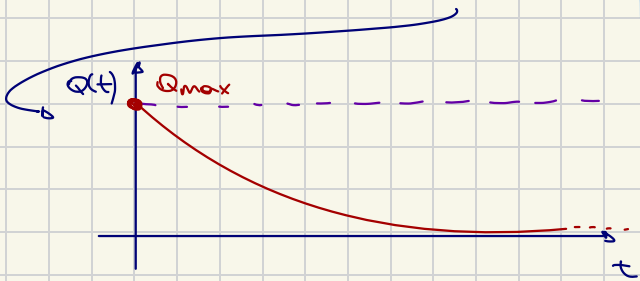
Scriviamo la II Legge di Kirchhoff:

$$\underbrace{-iR}_{\text{nel tratto AB}} + \underbrace{\frac{Q}{C}}_{\text{parte condensatore}} = 0$$

$\rightsquigarrow \quad iRC = Q \quad \rightsquigarrow \quad i(t) \cdot RC = Q(t)$ eq. differenziale

$Q'(t) \cdot R \cdot C = Q(t)$

$\rightsquigarrow$ Soluzione eq. diff. $\rightsquigarrow$ $Q(t) = \underbrace{Q_{max}}_{\text{carica iniziale del condensatore}} e^{-\frac{t}{RC}}$



Oss: Se avete caricato il condensatore con un gen. di tensione per la formula della carica vale $C f_{em} = Q_{max}$

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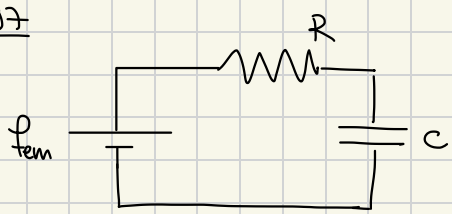

$f_{em} = 28{,}4\ V$

$R = 200\ k\Omega = 200 \cdot 10^3\ \Omega$

$C = 8{,}1\ \mu F = 8{,}1 \cdot 10^{-6} F$

Dopo $t = 2{,}76$ quanto vale Q?

$Q(t) = C f_{em} \left(1 - e^{-\frac{t}{RC}}\right) = \ldots = 1{,}88 \cdot 10^{-4} C$

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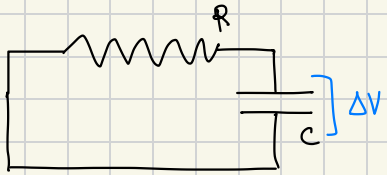

$\Delta V = 200V$ (dopo 0 s)

$\Delta V = 20V$ (dopo 30s)

- tempo caratteristico = ?
- ΔV dopo 40s

$$Q(t) = Q_{max} \cdot e^{-\frac{t}{RC}} \qquad C = \frac{Q}{\Delta V} \rightsquigarrow Q(t) = C\Delta V(t)$$

$$\Downarrow$$

$$\cancel{C}\,\underbrace{\Delta V(t)}_{\Delta V \text{ al tempo } t} = \cancel{C} \cdot \overbrace{\Delta V(0)}^{\Delta V \text{ al tempo } 0} \; e^{-\frac{t}{RC}}$$

$$\Delta V(30s) = \Delta V(0)\, e^{-\frac{30s}{RC}} \rightsquigarrow 20 = 200\, e^{-\frac{30}{RC}}$$

$$\frac{1}{10} = e^{-30/RC} \rightsquigarrow \frac{1}{10} = \frac{1}{e^{30/RC}}$$

$$\rightsquigarrow 10 = e^{30/RC} \rightsquigarrow \ln(10) = \frac{30}{RC} \rightsquigarrow \boxed{RC = \frac{30}{\ln(10)}\,s}$$

▷ Per calcolare $\Delta V(40s)$ basta sostituire 40s

$$\Delta V(40s) = \Delta V(0) \cdot e^{-\frac{40s}{RC}} =$$

$$= \Delta V(0)\, e^{-\frac{40}{30}\ln(10)}$$

$$= \Delta V(0)\, e^{-\frac{4}{3}\ln(10)} = \Delta V(0) \cdot 10^{-\frac{4}{3}} \approx 9{,}2\,V$$

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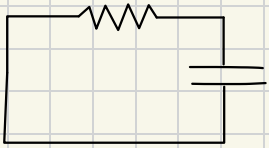

Q_{max} $\qquad \bar{t} = 1{,}42\,ms$

$Q(\bar{t}) = \frac{4}{10} Q_{max}$

Calcola $RC = t$ tempo caratteristico

$$Q(t) = Q_{max} \cdot e^{-\frac{t}{RC}}$$

$$\rightsquigarrow Q(\bar{t}) = \frac{4}{10} Q_{max} = Q_{max} \cdot e^{-\bar{t}/RC}$$

$$\frac{4}{10} = e^{-t/RC} \rightsquigarrow \ln\left(\frac{10}{4}\right) = \frac{t}{RC} \rightsquigarrow \boxed{RC = \frac{t}{\ln(10/4)}}$$

$$\frac{4}{10} = \frac{1}{e^{t/RC}}$$

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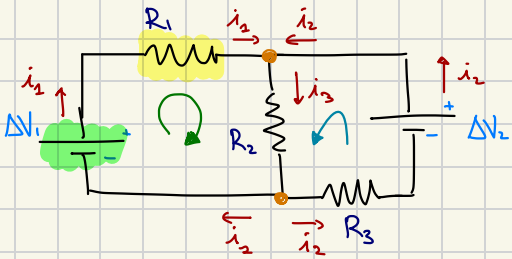

$\Delta V_1 = 12\,V$ $\quad R_1 = 10\,\Omega$

$\Delta V_2 = 24\,V$ $\quad R_2 = 20\,\Omega$

$R_3 = 30\,\Omega$

$$\begin{cases} i_1 + i_2 = i_3 \\ \Delta V_1 - i_1 R_1 - i_3 \cdot R_2 = 0 \\ \Delta V_2 - i_3 R_2 - i_2 R_3 = 0 \end{cases}$$

$$\begin{cases} i_1 + i_2 = i_3 \\ 12 - i_1 10 - 20 i_1 - 20 i_2 = 0 \\ 24 - 20(i_1 + i_2) - i_2 30 = 0 \end{cases}$$

$$\begin{cases} 30 i_1 + 20 i_2 = 12 \cdot 2 \\ 20 i_1 + 50 i_2 = 24 \cdot 3 \end{cases} \quad i_1 = x \quad i_2 = y$$

$$\begin{cases} 60x + 40y = 24 \\ 60x + 150y = 72 \end{cases} \quad \uparrow - \qquad 110y = 48 \qquad y = \frac{48}{110}$$

$$i_2 = \frac{48}{110}\,A \simeq 0{,}44\,A$$

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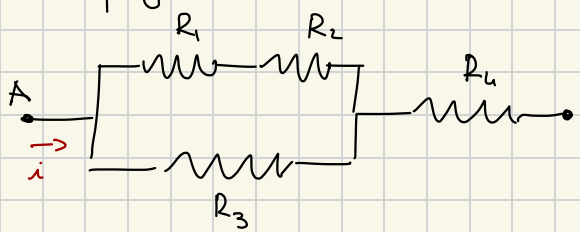

$R_2 = R_4 = R = ?$

$R_1 = 350\,\Omega$

$R_3 = 200\,\Omega$

$\Delta V_{AB} = 10\,V$

$i = 25\,mA$

$$R_{1,2} = R_1 + R_2 \qquad R_{1,2,3} = \frac{R_{1,2} \cdot R_3}{R_{1,2} + R_3} \qquad R_{1,2,3,4} = R_{1,2,3} + R_4$$

$$\Delta V_{AB} = i R_{1,2,3,4}$$

$$\Delta V_{AB} = i\left(R_{1,2,3} + R\right) = i\left(\frac{R_{1,2} \cdot R_3}{R_{1,2} + R_3} + R\right)$$

$$\Delta V_{AB} = i\left(\frac{(R_1 + R) \cdot R_3}{(R_1 + R) + R_3} + R\right)$$